



SRI INDU COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Batch:2023-2024

MINI PROJECT ON

OPTIMAL AMBULANCE POSITIONING FOR ROAD ACCIDENTS WITH
DEEP EMBEDDED CLUSTERING

INTERNAL GUIDE:

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OPTIMAL AMBULANCE POSITIONING FOR ROAD ACCIDENTS WITH DEEP EMBEDDED CLUSTERING



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ABSTRACT

- The number of casualties and fatalities brought on by road accidents is one of the most significant concerns in the modern world. Instead of dispatching ambulances only at the time of demand, pre-positioning them can reduce the response time and provide prompt medical attention.
- Deep learning techniques hold great potential and have proven to be essential for problem-solving and decision-making in the field of healthcare services.



INTRODUCTION

- Today one of the leading causes of death worldwide among children and adults is road accidents. The injuries caused by these fatal accidents cause considerable economic and personal losses to individuals, their families, and the country.
- An estimated 1.3 million individuals each year die as a result of road accidents. Between 20 and 50 million individuals experience non-fatal injuries, with many of them becoming disabled as a result.



- The ever-increasing growth in the number of automobiles is certain to have some negative consequences, the most likely of which is an increase in the frequency of fatal road accidents in densely populated places, resulting in a huge burden on the urban infrastructure.
- It is dreaded that if we fail to take definitive precautionary measures to overcome these statistics, then road accidents will take over as the fifth major cause of death by 2030.
- Despite these fatal consequences, this problem receives scanty attention and there is a lack of developing systematic methods to improve road safety.

LITERATURE SURVEY

- ▶ **TITLE:** A New Framework of Vehicle Collision Prediction by Combining SVM and HMM
- ▶ **AUTHOR:** Xiaoxia Xiong
- ▶ **ABSTRACT:** This paper presents a framework of accident prediction with a new perspective. First, the new framework of Chain of Road Traffic Incident (CRTI) is proposed, in which the observed vehicle movement features are viewed as road traffic system's external "performance" that, in essence, reflect the internal "health states" (safety states) of the system at a specific time. A two-stage modeling procedure of CRTI is then proposed using scenario-based strategy.

- ▶ **TITLE:** Survey on Enhancing Accident Safety: Technological Solutions
- ▶ **AUTHOR:** Vaishnavi Vhatkar¹, Mrs. Sonali Ajankar²
- ▶ **ABSTRACT:** Vehicle numbers are rising together with the population. Serious accidents that occur every day are rising on the road. This paper aims to bring forward an idea of where accidents can be prevented using methodologies that include IOT and deep learning concepts. Through an extensive review of the literature, it discusses a detailed comparative analysis of the current studies and research in this field.

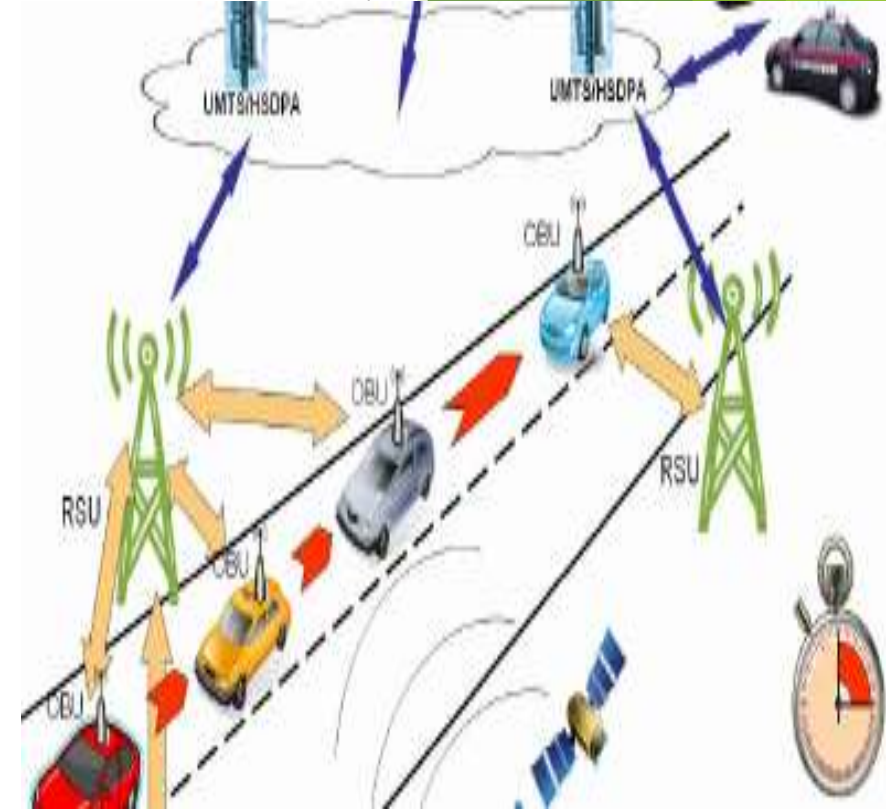
SYSTEM ANALYSIS

EXISTING SYSTEM

- Xiong et al. proposed Machine learning models for predicting accident vs non-accident patterns in crash sites using Gaussian Mixture models and SVM. They further predicted the severity of the crash injuries by clustering the crashes using fuzzy c-means, Feed Forward Neural Networks and SVM.

DISADVANTAGES

- The complexity of data
- Data availability
- Incorrect labeling



PROPOSED SYSTEM

- Exploratory Data Analysis (EDA) is performed on the real-time accident dataset through which the potential features and attributes which contribute towards accidents and patterns across the city are identified.
- A clustering-based approach using Deep Embedded Clustering (DEC) is developed to identify optimal ambulance positioning locations across Nairobi while preserving the feature relationships and patterns.

ADVANTAGES

The proposed system proposes an approach (optimal ambulance positioning framework) for the automatic placement of paramedic help using Deep Embedded Clustering (DEC).



SYSTEM REQUIREMENTS

FUNCTIONAL REQUIREMENTS

- ❖ RESPONSE CORDINATOR
- ❖ USER PORTAL



NON-FUNCTIONAL REQUIREMENTS

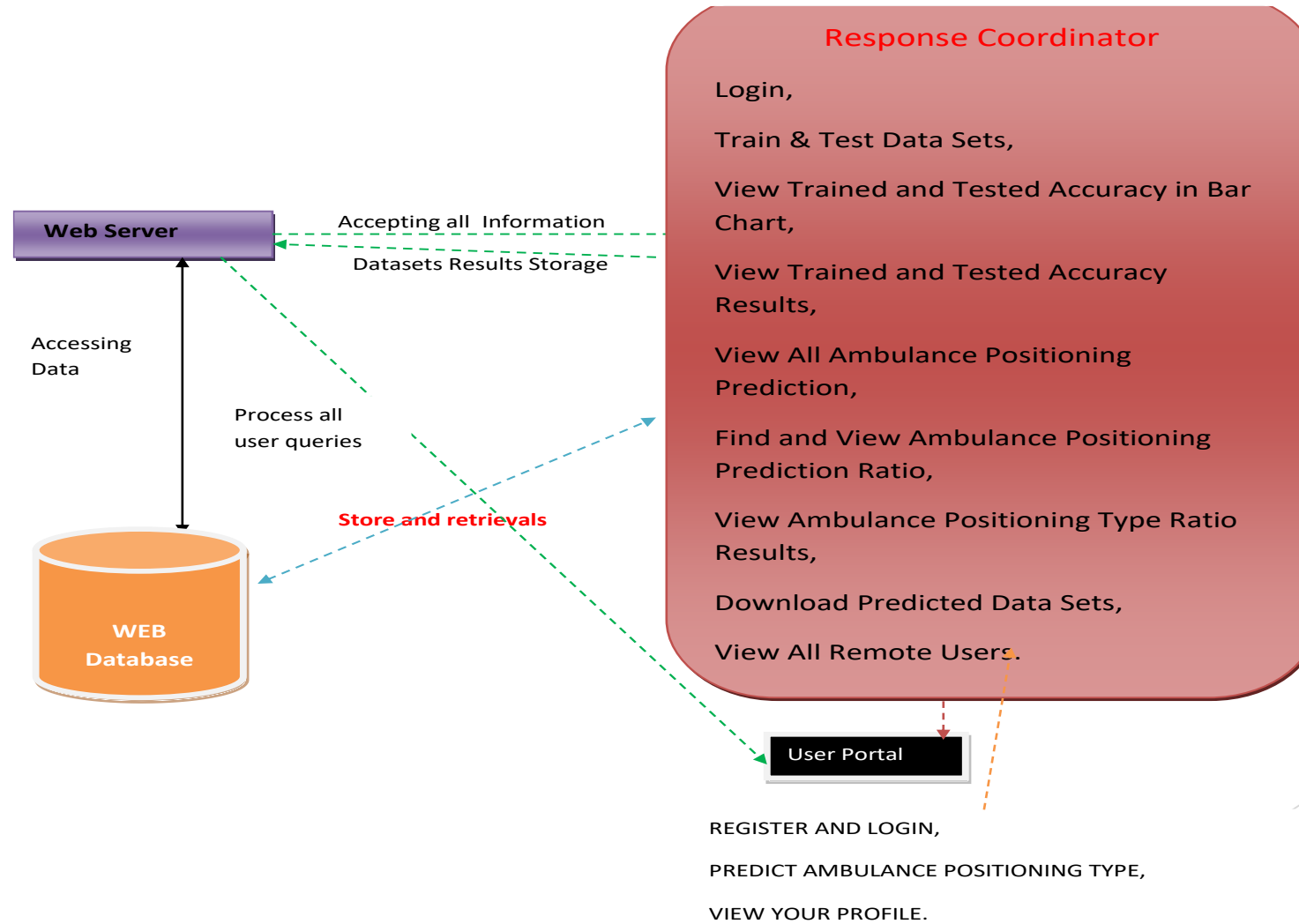
HARDWARE REQUIREMENTS

1. Processor - Pentium –IV
2. RAM - 4 GB (min)
3. Hard Disk - 20 GB
4. Key-Board - Standard Windows Keyboard
5. Mouse - Two or Three Button Mouse
6. Monitor - SVGA

SOFTWARE REQUIREMENTS

1. Operating system : Windows 7 Ultimate.
2. Coding Language : Python.
3. Front-End : Python.
4. Back-End : Django-ORM
5. Designing : Html, CSS, JavaScript.
6. Data Base : MySQL (WAMP Server).

SYSTEM ARCHITECTURE

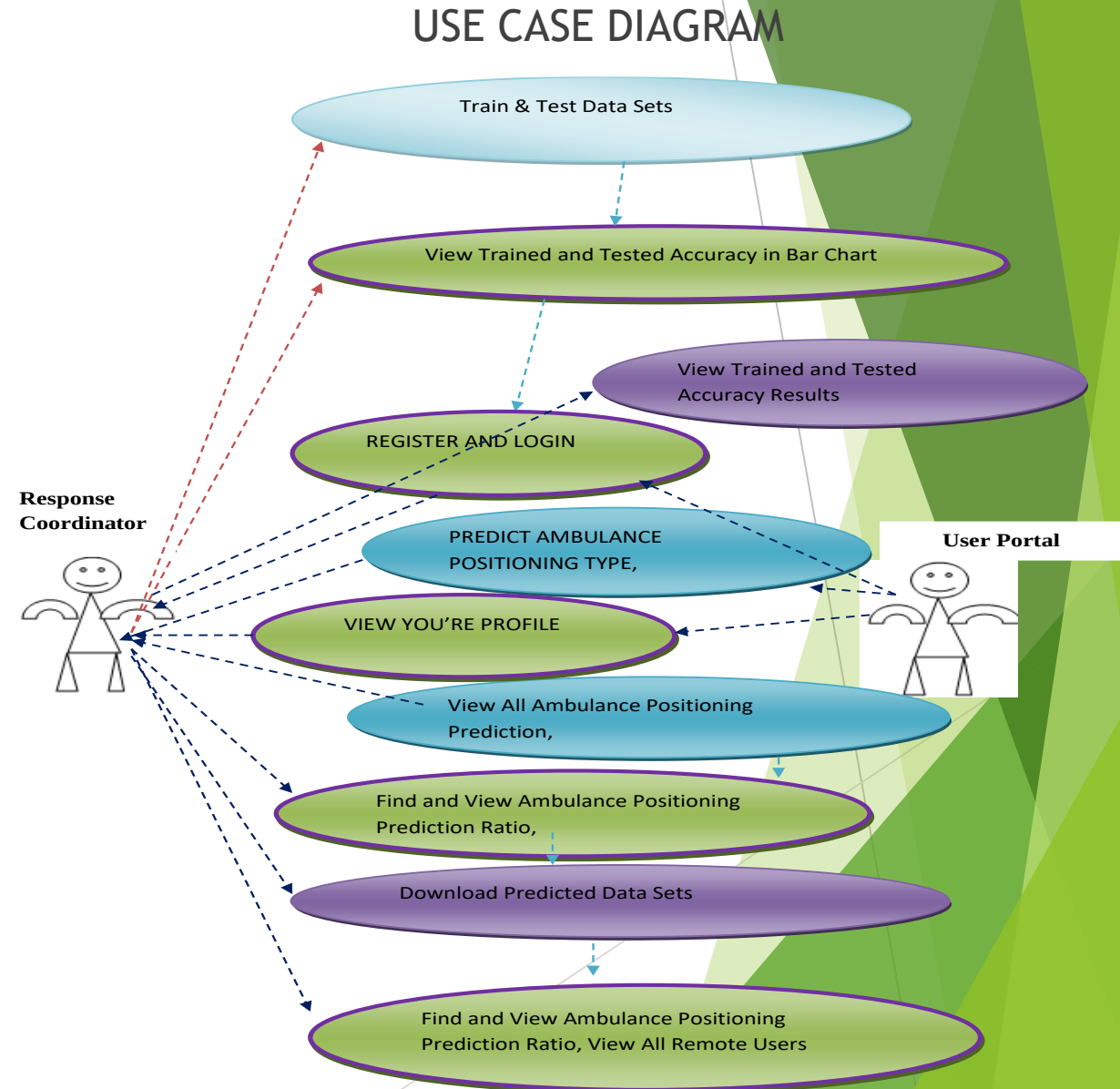


UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering.

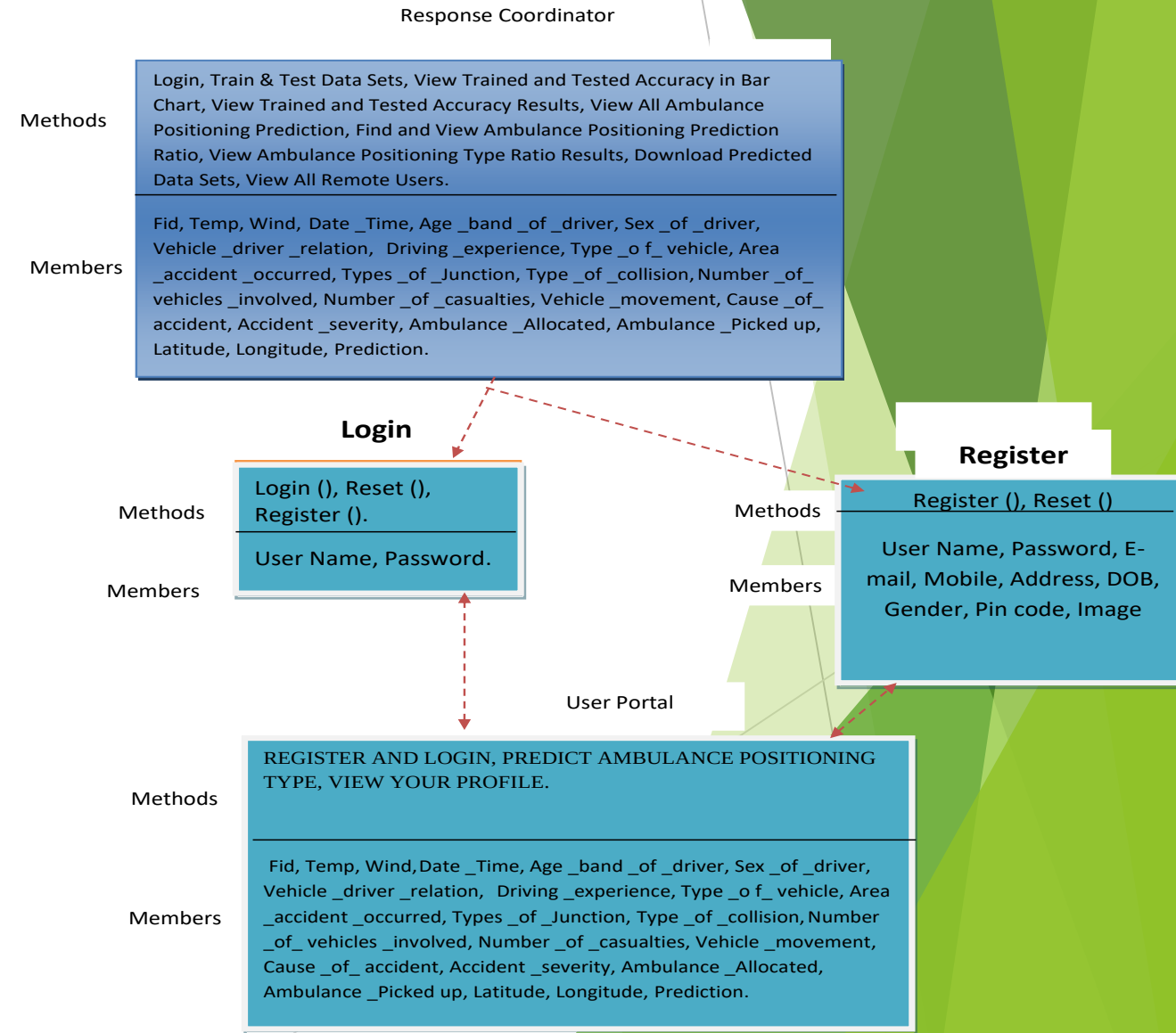
❖ USE CASE DIAGRAM

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis.



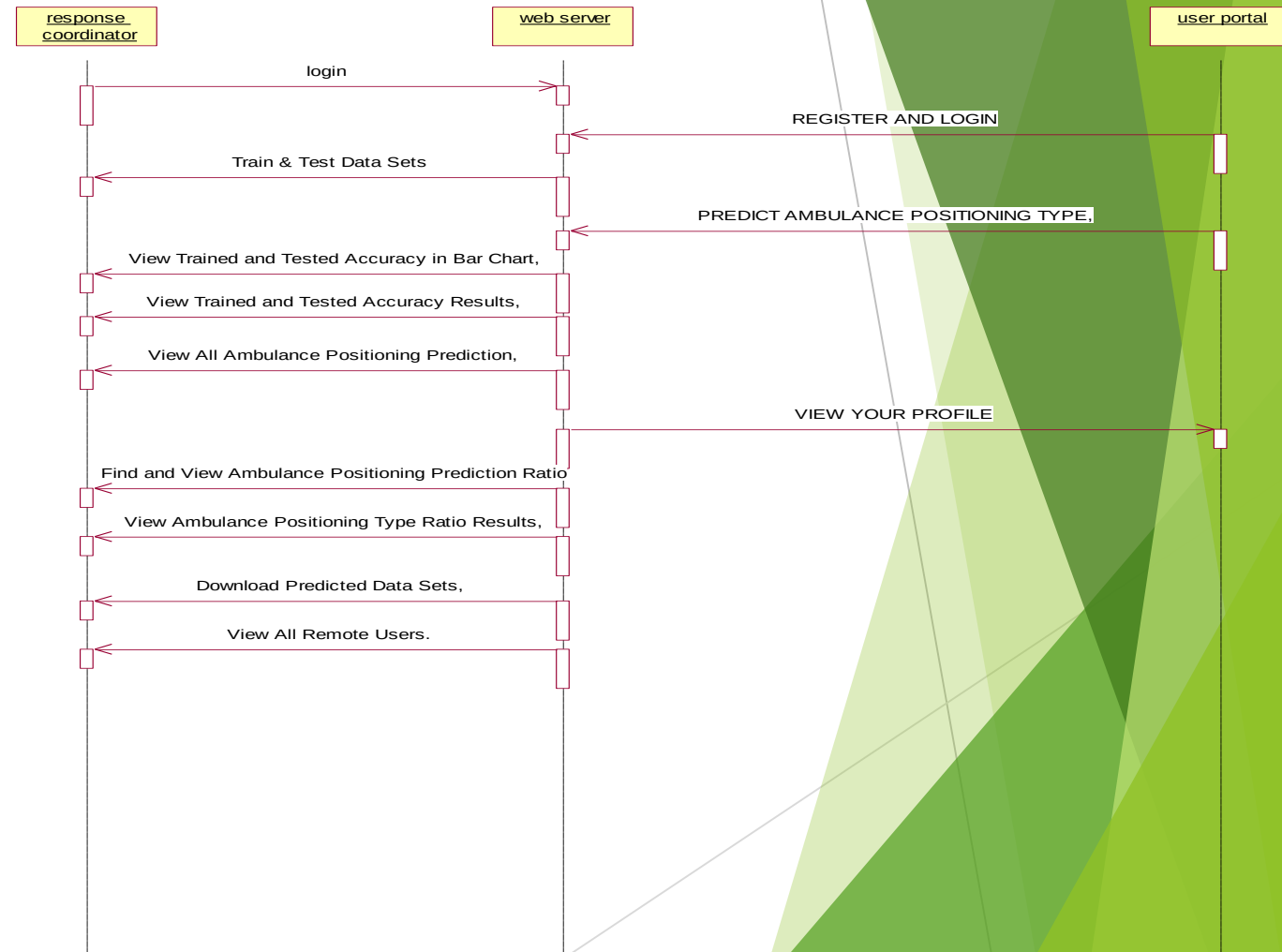
CLASS DIAGRAM

- In software engineering, a class diagram in the Unified Modeling Language is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations and the relationships among the classes.



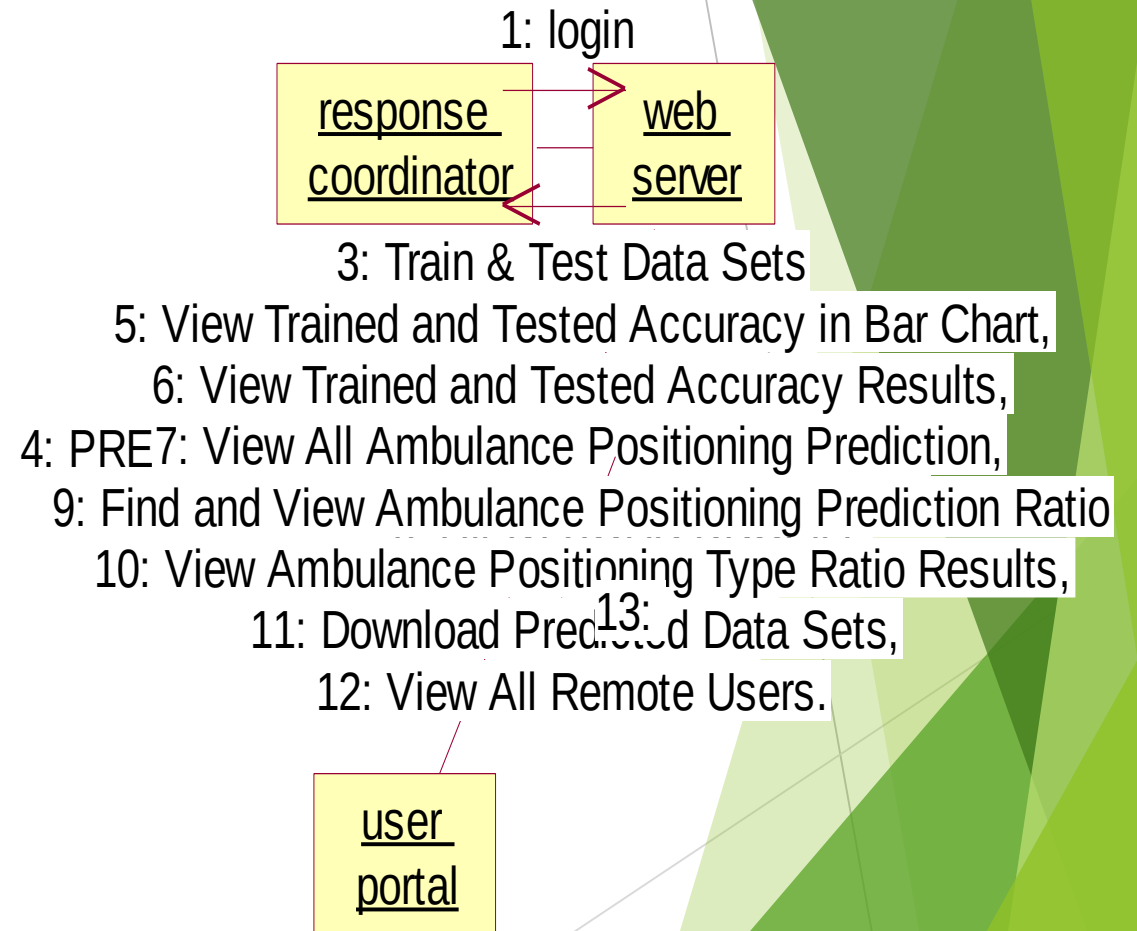
SEQUENCE DIAGRAM

- A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart.

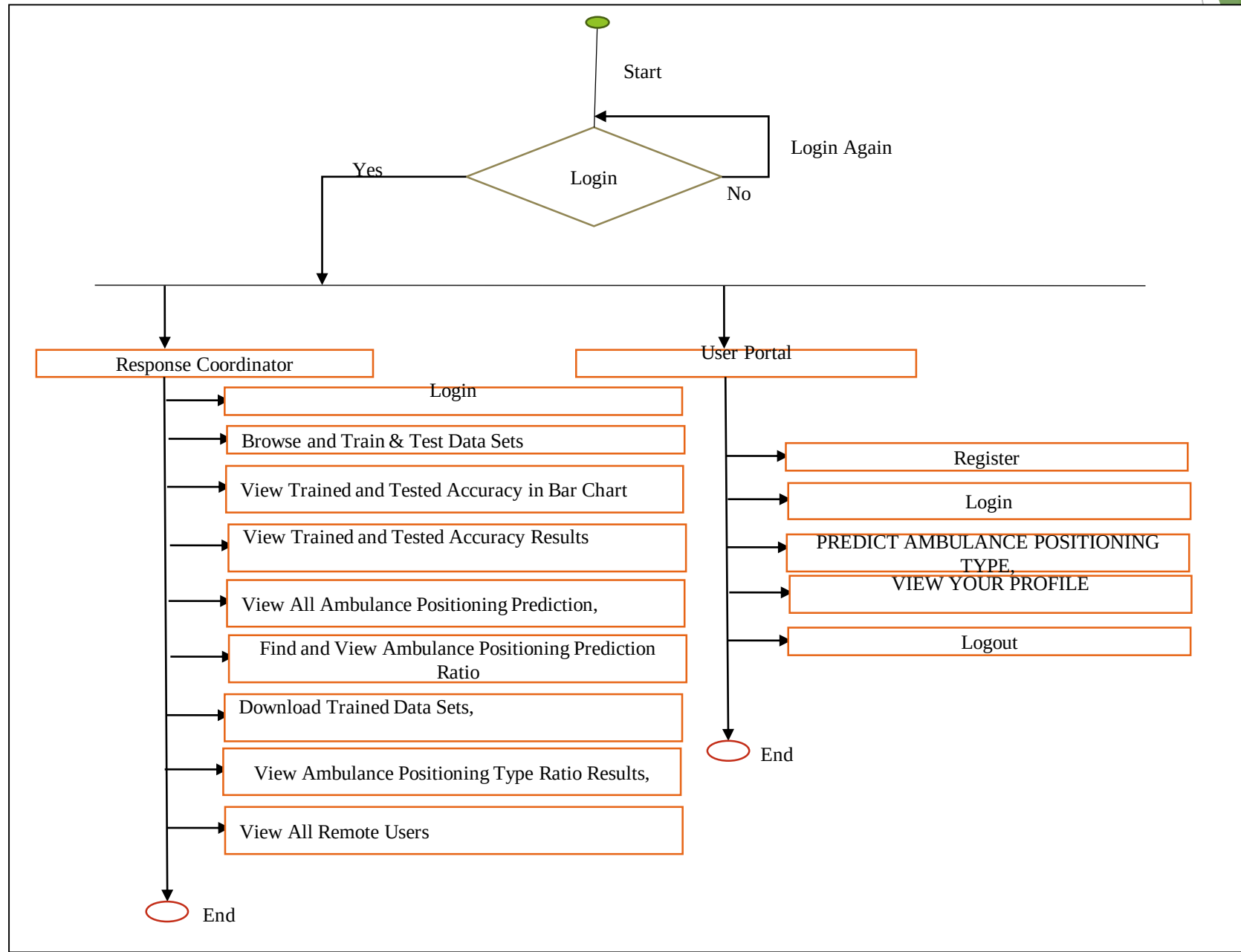


COLLABRATION DIAGRAM

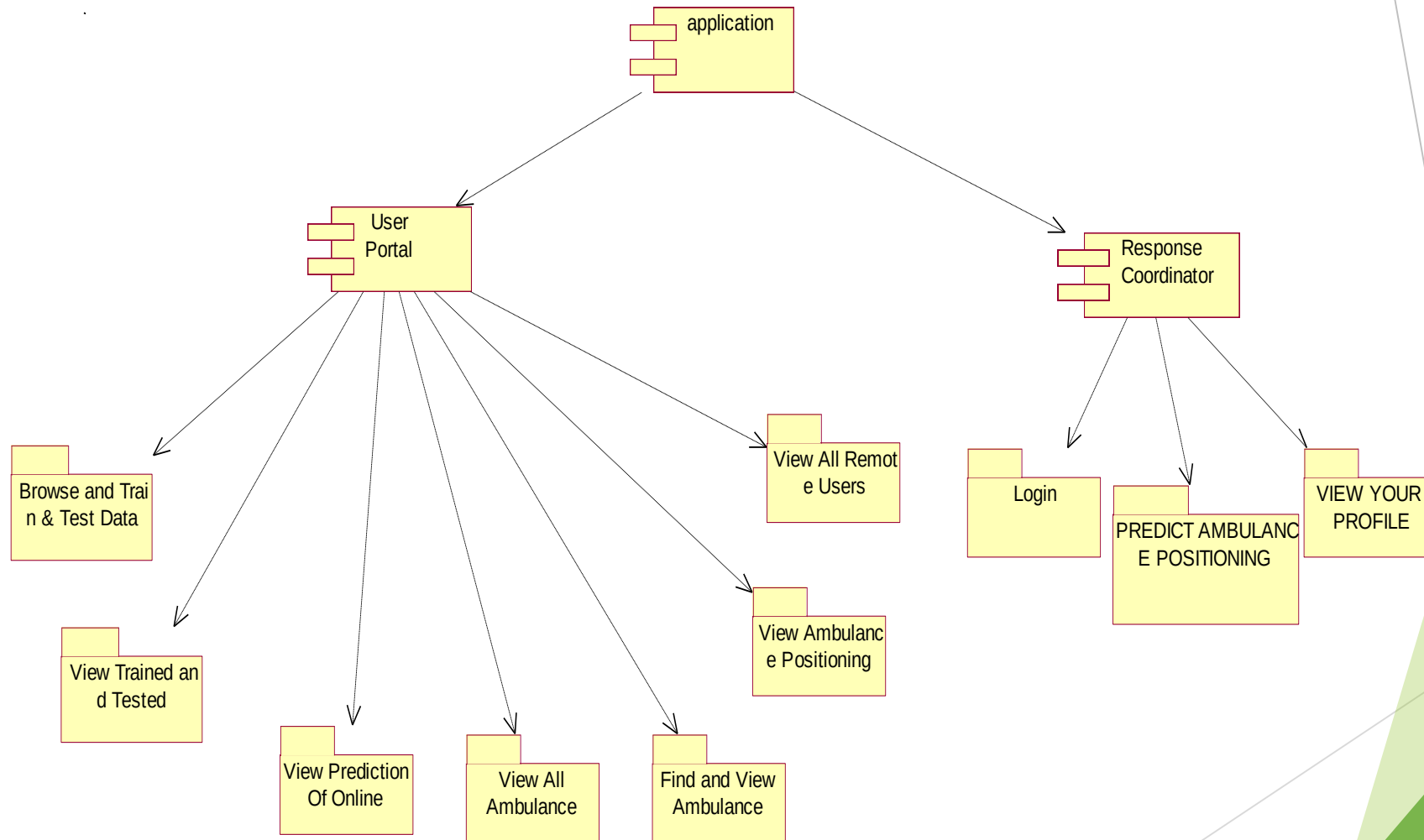
- Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency.



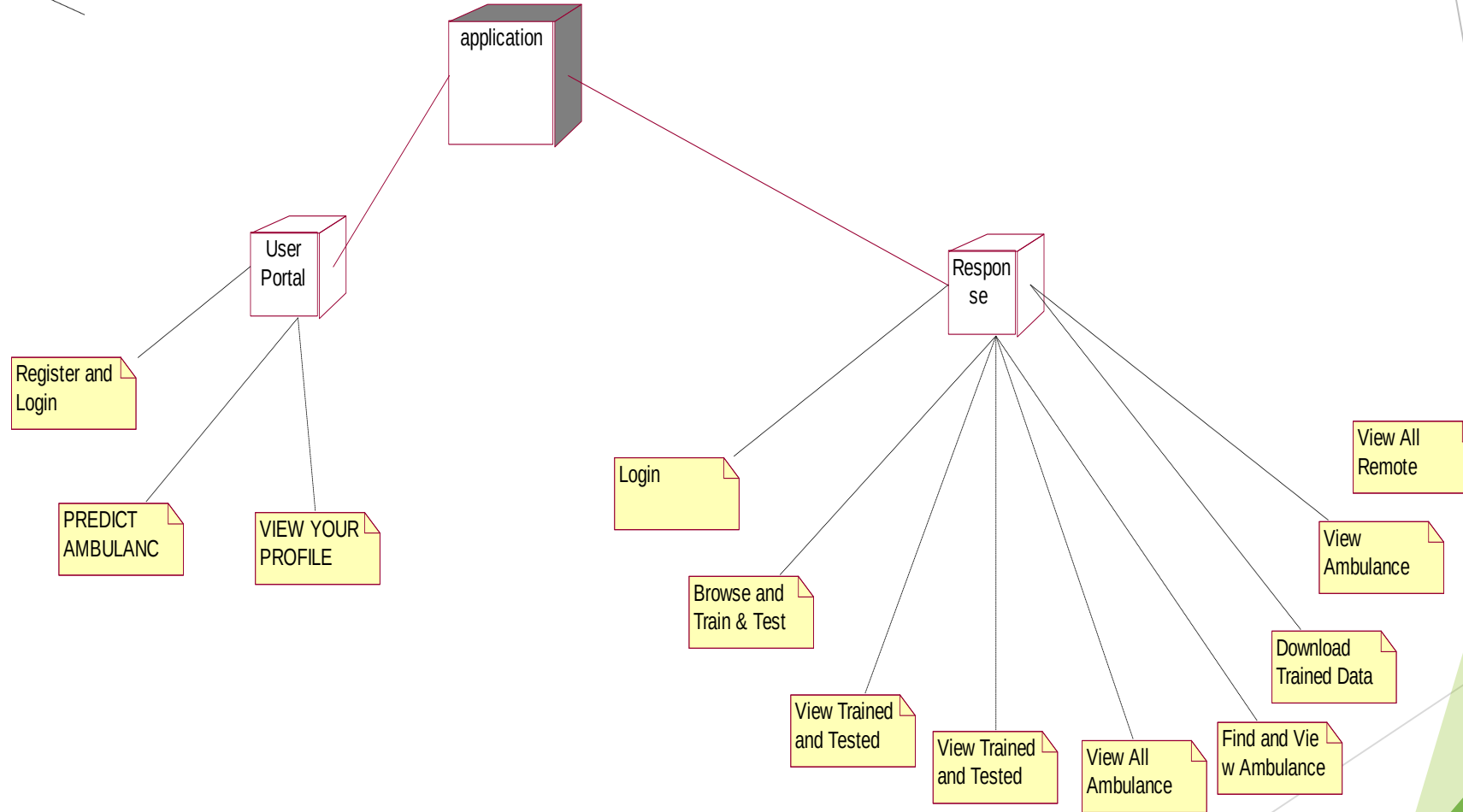
ACTIVITY DIAGRAM



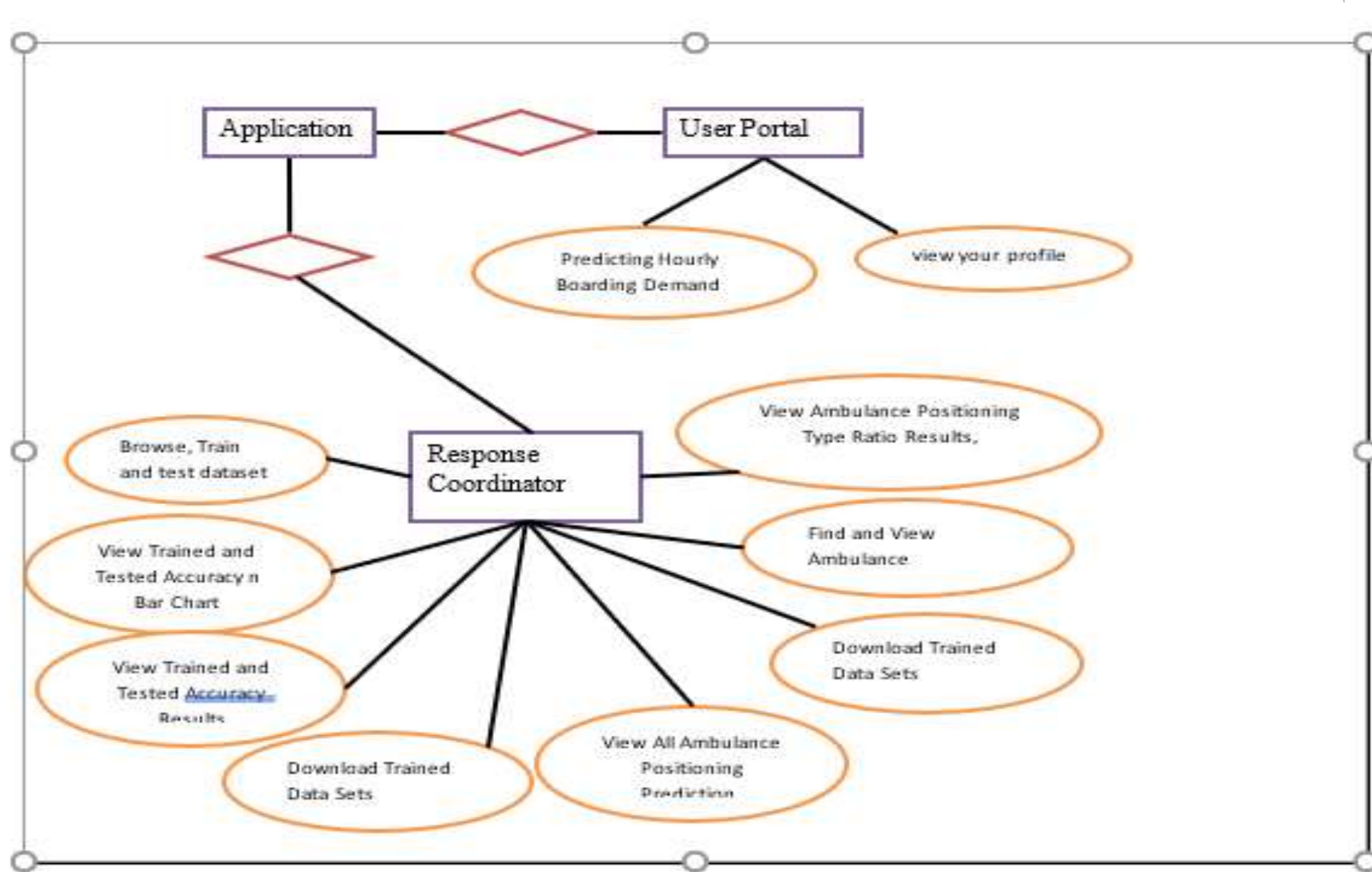
COMPONENT DIAGRAM



DEPLOYMENT DIAGRAM



ER DIAGRAM



IMPLEMENTATION

MODULES

- Response Coordinator
- User Portal

MODULE DESCRIPTION

➤ RESPONSE COORDINATOR

- In this module, the Service Provider has to login by using valid user-name and password. After login successful he can do some operations such as Train & Test Data Sets, View Trained and Tested Accuracy in Bar Chart, etc.

➤ USER PORTAL

- In this module, there are n numbers of users are present. User should register before doing any operations. Once user registers, their details will be stored to the database.

SAMPLE CODE

Manage.py

```
#!/usr/bin/env python

"""Django's command-line utility for administrative tasks."""

import os

import sys

def main():

    """Run administrative tasks."""

    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'optimal_ambulance_positioning.settings')

    try:

        from django.core.management import execute_from_command_line

    except ImportError as exc:

        raise ImportError(

            "Couldn't import Django. Are you sure it's installed and "

            "available on your PYTHONPATH environment variable? Did you "

            "forget to activate a virtual environment?"

        ) from exc

    execute_from_command_line(sys.argv)

if __name__ == '__main__':

    main()
```

PREDICT.HTML

```
{% extends 'RUser/design.html' %}

{% block userblock %}

<link rel="icon" href="images/icon.png" type="image/x-icon" />

<link href="https://fonts.googleapis.com/css?family=Lobster" rel="stylesheet">

<link href="https://fonts.googleapis.com/css?family=Righteous" rel="stylesheet">

<link href="https://fonts. whitegoogleapis.com/css?family=Fredoka+One" rel="stylesheet">

<style>

body {background-color:#000000;}

.container-fluid {padding:50px;}

.container{background-color:;padding:50px; }

#title{font-family: 'Fredoka One', cursive;

}

.text-uppercase{

font-family: 'Righteous', cursive;

}

.tweettext{

border: 2px solid yellowgreen;

width: 1104px;

height: 142px;
```




```
.style1 {
```

```
    color: #FF0000;
```

```
    font-weight: bold;
```

```
}
```

```
.style6 {color: #FF0000}
```

```
.style8 {font-weight: bold}
```

```
.style10 {color: #FFFF00}
```

```
    .style12 {color: #FFFF00; font-weight: bold; }
```

```
    .style13 {
```

```
        color: #FFFF00;
```

```
        font-weight: bold;
```

```
        font-family: Georgia, "Times New Roman", Times, serif;
```

```
        font-style: italic;
```

```
}
```

```
</style>
```

```
<body>
```

```
<div class="container-fluid">
```

```
    <div class="container">
```

```
<div class="row">
```

```
    <div class="col-md-5">
```

```
<form role="form" method="POST" >
```

```
{% csrf_token %}
```

```
<fieldset>
```

```
    <p align="center" class="text-uppercase pull-center style1">PREDICT AMBULANCE POSITIONING  
    TYPE!!! </p>
```

```
<hr>
```

```
{% csrf_token %}
```

```
<table width="1122" align="center">
```

```
<tr>
```

```
    <td height="44" colspan="4" bgcolor="#FF0000"><div align="center" class="style13">  
    ENTER ALL PROFILE ATTRIBUTE DETAILS HERE !!! </div></td>
```

```
</tr>
```

```
<tr>
```

```
    <td width="289" height="44" bgcolor="#FF0000"><div align="center"><strong><span  
    class="style10"><br>
```

```
    Enter Fid</span></strong></div></td>
```

```
<td width="240"><div align="center">
```

```
    <input type="text" name="Fid">
```

```
</div></td>
```

```
<td><div align="center">

    <input name="Driving_experience" type="text" value="">

</div></td>

</tr>

<tr>

    <td height="44" bgcolor="#FF0000"><div align="center"><strong><span class="style10">Enter
Type_of_vehicle</span></strong></div></td>

    <td><div align="center">

        <input type="text" name="Type_of_vehicle">

    </div></td>

<td bgcolor="#FF0000"><span class="style12">Enter Area_accident_occured</span></td>

    <td><div align="center">

        <input type="text" name="Area_accident_occured">

    </div></td>

</tr>

</tr>

<tr>

    <td height="44" bgcolor="#FF0000"><div align="center"><strong><span class="style10">Enter
Types_of_Junction</span></strong></div></td>

    <td><div align="center">

        <input type="text" name="Types_of_Junction">

    </div></td>
```

```
<td bgcolor="#FF0000"><span class="style12">Enter Type_of_collision</span></td>

    <td><div align="center">

        <input type="text" name="Type_of_collision">

    </div></td>

</tr>

<tr>

    <td height="44" bgcolor="#FF0000"><div align="center"><strong><span
class="style10">Enter Number_of_vehicles_involved</span></strong></div></td>

    <td><div align="center">

        <input name="Number_of_vehicles_involved" type="text" value="">

    </div></td>

<td bgcolor="#FF0000"><span class="style12">Enter Number_of_casualties</span></td>

    <td><div align="center">

        <input type="text" name="Number_of_casualties">

    </div></td>

</tr>

<tr>
```

```
{% extends 'RUser/design.html' %}

{% block userblock %}

<link rel="icon" href="images/icon.png" type="image/x-icon" />

<link href="https://fonts.googleapis.com/css?family=Lobster" rel="stylesheet">

<link href="https://fonts.googleapis.com/css?family=Righteous" rel="stylesheet">

<link href="https://fonts.googleapis.com/css?family=Fredoka+One" rel="stylesheet">

<style>

body {background-color:#000000;}

.container-fluid {padding:50px;}

.container{background-color:white;padding:50px;

}

#title{font-family: 'Fredoka One', cursive;

}

.text-uppercase{

font-family: 'Righteous', cursive;

}

.style1 {

color: #FF0000;

font-weight: bold;

}
```

```
</div>

<div class="form-group">

    <p><span class="style1">COUNTRY</span> = {{object.country}}</p>

</div>

    <div class="form-group">

        <p><span class="style1">STATE</span> = {{object.state}}</p>

    </div>

        <div class="form-group">

            <p><span class="style1">CITY</span> = {{object.city}}</p>

        </div>

    </fieldset>

</form>

</div>

    <div class="col-md-2">

<!--null-->

</div>

        </div>

        </div>

        </div>

{% endblock %}
```

OUTPUT SCREENS



Optimal Ambulance Positioning for Road Accidents With Deep Embedded Clustering

[Train & Test DataSets](#) [View Trained and Tested Accuracy in Bar Chart](#) [View Trained and Tested Accuracy Results](#) [View All Ambulance Positioning Prediction](#)

[Find and View Ambulance Positioning Prediction Ratio](#) [View Ambulance Positioning Type Ratio Results](#) [Download Predicted Data Sets](#) [View All Remote Users](#) [Logout](#)

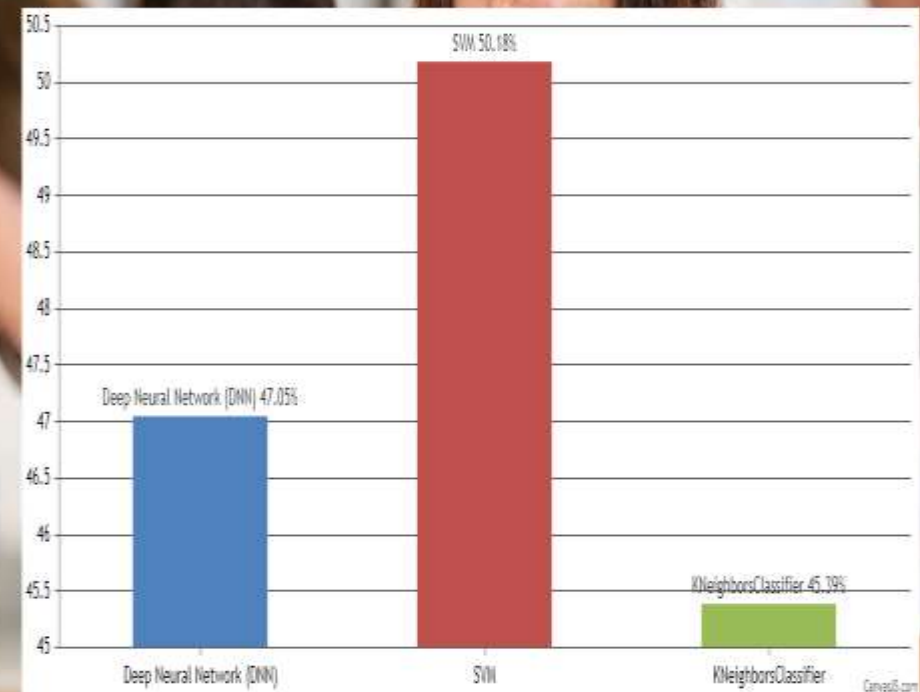
VIEW ALL REMOTE USERS !!!

USER NAME	EMAIL	Mob No	Country	State	City
Gopal	Gopal123@gmail.com	9535866270	India	Karnataka	Bangalore
Manjunath	tmksmanju19@gmail.com	9535866270	India	karnataka	Bangalore

Optimal Ambulance Positioning for Road Accidents With Deep Embedded Clustering

[Train & Test DataSets](#) [View Trained and Tested Accuracy in Bar Chart](#) [View Trained and Tested Accuracy Results](#) [View All Ambulance Positioning Prediction](#)

[Find and View Ambulance Positioning Prediction Ratio](#) [View Ambulance Positioning Type Ratio Results](#) [Download Predicted Data Sets](#) [View All Remote Users](#) [Logout](#)



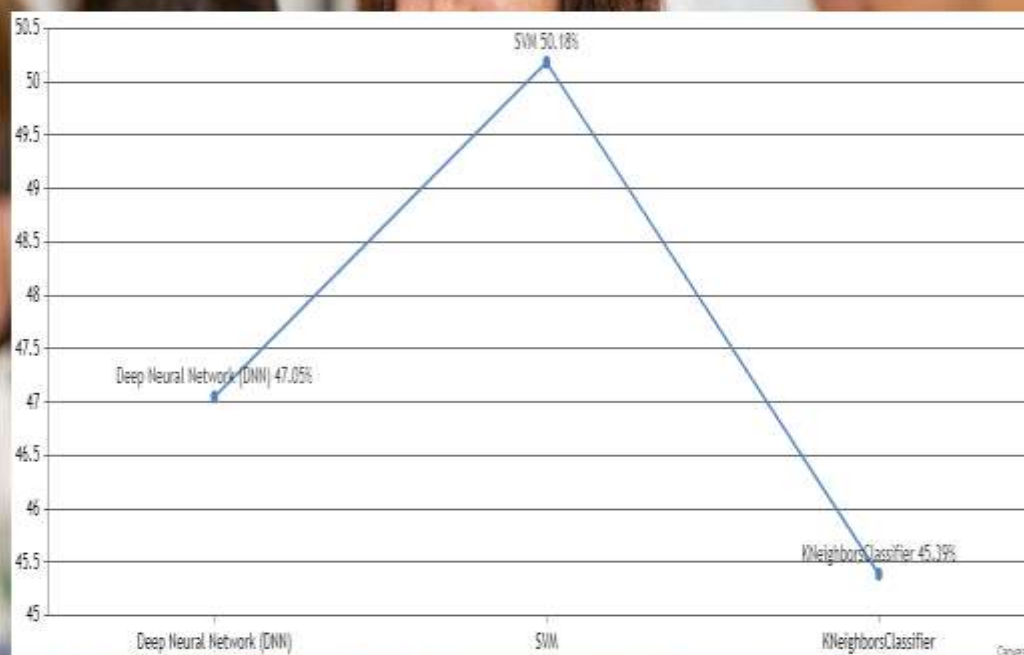
Optimal Ambulance Positioning for Road Accidents With Deep Embedded Clustering

[Train & Test DataSets](#) [View Trained and Tested Accuracy in Bar Chart](#) [View Trained and Tested Accuracy Results](#) [View All Ambulance Positioning Prediction](#)

[Find and View Ambulance Positioning Prediction Ratio](#) [View Ambulance Positioning Type Ratio Results](#) [Download Predicted Data Sets](#) [View All Remote Users](#) [Logout](#)

LINE CHART

LINE CHART



Optimal Ambulance Positioning for Road Accidents With Deep Embedded Clustering

[Train & Test DataSets](#) [View Trained and Tested Accuracy in Bar Chart](#) [View Trained and Tested Accuracy Results](#) [View All Ambulance Positioning Prediction](#)

[Find and View Ambulance Positioning Prediction Ratio](#) [View Ambulance Positioning Type Ratio Results](#) [Download Predicted Data Sets](#) [View All Remote Users](#) [Logout](#)

VIEW ALL REMOTE USERS !!!

USER NAME	EMAIL	Mob No	Country	State	City
Gopal	Gopal123@gmail.com	9535866270	India	Karnataka	Bangalore
Manjunath	tmksmanju19@gmail.com	9535866270	India	karnataka	Bangalore

Clustering

Deep embedded clustering (DEC), C₁2Vec, k-means, ambulance positioning, accident hotspots



REGISTER YOUR DETAILS HERE !!!

sign_up

CONCLUSION

- ▶ Over the past 20 years, methods for identifying accident hotspots and determining optimal paramedic positions have evolved and now plays a significant role in the successful implementation of traffic safety management programs.
- ▶ This study aimed to develop and compare models for predicting optimal locations for positioning ambulances in Nairobi city, based on the Nairobi accidents dataset from 2018 to 2019.
- ▶ Following the data preprocessing and feature selection, a clustering-based approach was followed using Deep Embedded Clustering along with standard machine learning algorithms like K-Means clustering, GMM, and Agglomerative clustering to identify five clusters, the centroids of which provided the optimal ambulance positions

FUTURE SCOPE

- The future scope of "Optimal Ambulance Positioning for Road Accidents with Deep Embedded Clustering" is highly promising and multifaceted.
- Advances in machine learning and artificial intelligence, particularly in deep learning, offer the potential to significantly enhance the precision and efficiency of ambulance deployment.
- Additionally, the integration of predictive analytics could foresee potential accident locations based on environmental and temporal factors, further refining positioning strategies.



REFERENCES

- ❖ A. J. Ghandour , H. Hammoud, and S. Al-Hajj, “Analyzing factors associated with fatal road crashes: A machine learning approach,” *Int. J. Environ. Res. Public Health*, vol. 17, no. 11, p. 4111, Jun. 2020.
- ▶
- ❖ P. Tiwari, H. Dao, and N. G. Nguyen, “Performance evaluation of lazy, decision tree classifier and multilayer perceptron on traffic accident analysis,” *Informatica*, vol. 41, no. 1, pp. 39–46, 2017.
- ▶
- ❖ T. A. Granberg and H. T. N. Nguyen, “Simulation based prediction of the near-future emergency medical services system state,” in *Proc. Winter Simul. Conf. (WSC)*, Dec. 2018, pp. 2542–2553.
- ▶

Thank You!